CAP-01 / DOCUMENT 01

Project Planning and Management

Shopping Copilot

Arabic-first shopping advice and guarded Storefront actions

Programme	Digital Egypt Pioneers Initiative (DEPI)
Track	AWS ML Engineering
Submission deadline	16 October 2026
Export date	16 October 2026
Source document date	16 October 2026
Release status	Prepared for submission; official upload pending

Prepared for the AWS ML Engineering graduation submission. The owner confirms lecturer approval and the organization-repository invitation were received. Approval wording/date and numerical grading details were not supplied. Assigned responsibilities are distinguished from actual contributions.

Project planning and management

Project: Shopping Copilot - Arabic-first shopping advice and guarded Storefront actions.

Updated: 16 October 2026. **Track:** AWS ML Engineering. **Published DEPI deadline:** 16 October 2026. **Status:** prepared for submission under the supplied guidelines. The owner confirms lecturer approval and the repository invitation were received; no dated approval reference was supplied. Official upload and numerical grading weights remain unrecorded.

Actual working arrangement (2 October 2026): the owner and coding assistant perform the work; the owner coordinates this deliverable. Earlier named academic leads were administrative allocations, not verified contributions or available staffing. See the updated CAP-01 register.

Purpose and objectives

Shopping Copilot helps a Shopper describe a shopping need, compare products and complete shopping tasks through natural conversation. It focuses on Egyptian Arabic, Franco-Arabic, English and mixed-language input on one Controlled Storefront. The goal is to combine useful decision support with visible, bounded actions: the language model interprets requests, while deterministic controls govern what the application may execute.

The graduation objectives are to:

1. Demonstrate catalogue-grounded advice and the supported shopping journeys without giving the model unrestricted browser authority.
2. Evaluate intent interpretation, task outcomes and runtime safety separately, preserving failed cases and measurement limitations.
3. Build a reviewed multilingual dataset and compare an offline TF-IDF + Logistic Regression baseline against the LLM on the same intent-classification task.
4. Deliver the Controlled Storefront application on AWS after model, security, budget and deployment acceptance checks.
5. Produce traceable documentation, individual contribution records and a reproducible final demonstration.

Scope and current baseline

The local MVP is tagged **mvp-1** at **39ac6d1**. It includes a FastAPI Agent, TypeScript Bridge and Panel, and a Controlled Storefront with fictional products, accounts, cart and checkout. It supports discovery, conversational advice, navigation, Spotlight, reversible cart changes, Undo and bound Confirmation for guarded mutations.

RESULTS.md records the acceptance basis. Historical full live runs passed 43/44, 44/44 and 44/44 tasks, with 19/19 designated safety cases per run on their frozen runtime. The final local release passed 458 Python and 155 TypeScript tests. The owner explicitly accepted the informal human-study scope and historical full live results plus current targeted/automated verification for this release. These are distinct evidence sources, not a claim that a new full live gate ran on the tagged advice implementation.

AWS deployment, independent unseen-data evaluation and the final same-input LLM comparison remain planned. Offline classical-model training and a seven-family exploratory comparison were completed on 7 October, as summarized below. Independent retailer compatibility, Shopify/WooCommerce integrations, real

payments, broad multi-tenancy, SaaS billing and production-readiness claims are outside the graduation scope. No new infrastructure or paid experiment is authorized by this planning document.

ML evidence available at this submission

The seven-family comparison used 210 training and 53 validation examples, all synthetic and exposed, with zero unseen examples. All 46 planned candidate fits converged. Frozen multilingual embeddings plus Logistic Regression achieved the highest selected validation accuracy, 23/53 (43.40%), with macro-F1 0.4304; TF-IDF plus Logistic Regression scored 20/53 (37.74%), with macro-F1 0.3776. These are weak exploratory classification results, not independent test accuracy or an LLM comparison. The executed notebook records the full experiment, errors, confusion matrices, timings and provenance. Classifiers remain offline and never execute shopping Actions.

CAP-02 and CAP-03 remain open for independent evaluation and the final fair LLM comparison. The next experiment needs a feasible independent data/label-review arrangement; further tuning of the reused validation set cannot establish generalisation.

Team and responsibility boundaries

Member: Ahmed Yasser

Assigned responsibility: AI Agent and LLM architecture

Main evidence area: Provider contract, prompting, Structured Intent, deterministic safety boundaries.

Member: Ali Amr

Assigned responsibility: AWS cloud architecture and infrastructure

Main evidence area: Bedrock integration, cloud design, budget monitoring and deployment security.

Member: Mohamed Hamdi

Assigned responsibility: Backend and Agent orchestration

Main evidence area: Sessions, Server-Sent Events, action lifecycle and recovery reconciliation.

Member: Ezz Mohamed

Assigned responsibility: Frontend and Storefront integration

Main evidence area: Bridge, Panel, cross-origin communication, responsive layout and RTL.

Member: Rana Ali

Assigned responsibility: QA, safety and evaluation

Main evidence area: Playwright tests, multilingual benchmarking and safety reports.

These are the owner's supplied technical assignments. They do not retroactively attribute every existing commit to a member. The owner coordinates the overall submission; only the owner and coding assistant actively perform the work. There is no available independent human annotation reviewer. The supplied guidelines specify organization-owned GitHub delivery; the owner supplied the submission repository on 16 October. No editable report template was supplied. Emails and participant identities are excluded from public evidence.

Work breakdown and delivery sequence

Work package: Local MVP

Deliverable and exit condition: Versioned application and honest technical/user results

Dependency / status: Closed under documented adjustments.

Work package: CAP-01

Deliverable and exit condition: Confirmed requirements, responsibility/deadline register, planning and literature reports mapped to submission requirements

Dependency / status: Reports prepared; approval/invitation owner-confirmed; upload record pending.

Work package: CAP-02

Deliverable and exit condition: Reviewed dataset, annotation guidance, provenance, grouped splits and protected unseen evaluation set

Dependency / status: Synthetic development releases exist; independent evaluation release remains open.

Work package: CAP-03

Deliverable and exit condition: Reproducible offline baseline, fair LLM comparison and error analysis

Dependency / status: Exploratory offline fits completed; independent test and final LLM comparison pending.

Work package: CAP-04

Deliverable and exit condition: Reviewed AWS architecture, threat boundaries, restart behaviour and cost plan

Dependency / status: Can proceed alongside ML work after local closure; spending approval remains separate.

Work package: CAP-05

Deliverable and exit condition: Bedrock adapter and model qualification

Dependency / status: Requires data/comparison/design evidence.

Work package: CAP-06-07

Deliverable and exit condition: Repeatable deployment, rollback and cloud acceptance results

Dependency / status: Qualified model and approved cloud design/budget.

Work package: CAP-08

Deliverable and exit condition: Final report, presentation, demo and member-level evidence

Dependency / status: Documentation maintained throughout; final closure follows technical acceptance.

The classical baseline remains offline. It neither replaces the shopping interpreter nor receives permission to execute Actions. CAP-02 and CAP-03 are separate from the existing exposed regression corpus; prior development examples cannot become an unseen evaluation set by renaming them.

Confirmed deadlines and proposed checkpoints

Confirmed DEPI deliverable	Deadline
Planning and management; literature review; requirements gathering	16 October 2026
System analysis and design	6 November 2026
Implementation, source code and execution	30 November 2026
Final presentation, testing and reports	4 December 2026

For the first deadline, the proposed internal sequence is content and source review by 9 October, team/format review by 13 October, and export/link/readiness checks by 15 October. These are planning proposals, not instructor-issued dates or commitments from named members. Plan actual work around the owner and coding assistant; do not depend on nominal team staffing. Detailed CAP-02-07 estimates depend on annotation capacity, cloud access and the approved budget; do not compress qualification or fabricate results to fit a date.

Proposed Gantt and resources

The work windows below are planning estimates as of 2 October, not evidence that work is complete. Official due dates are milestones. CAP-03 cannot start before reviewed CAP-02 data; CAP-05-07 cannot bypass qualification or budget approval if an estimated window slips.

Proposed graduation work windows

Planning estimates; official deadlines are diamonds

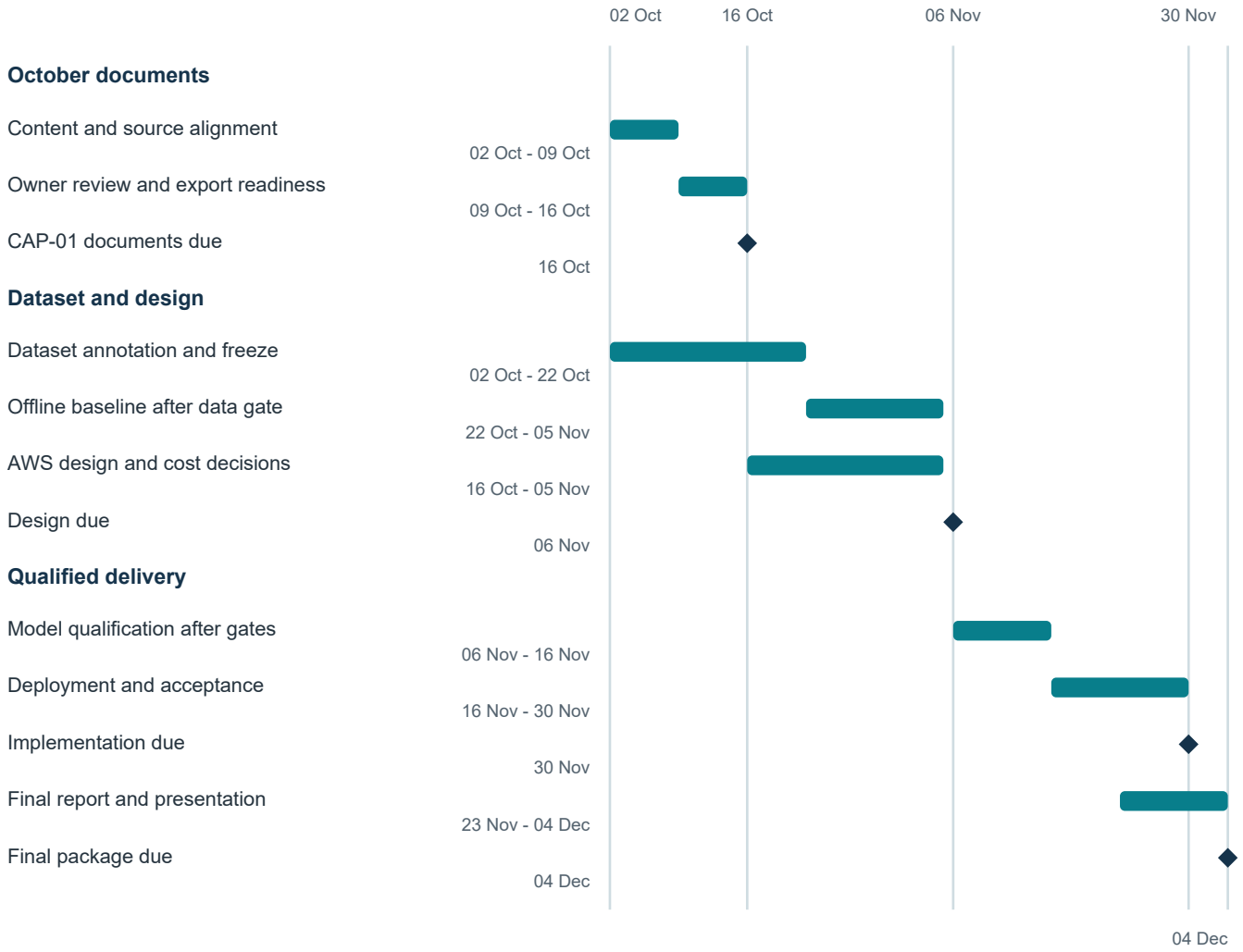


Figure 1. Proposed work windows from the planning source. End dates are derived from the stated duration; gates remain binding if windows slip.

Resource	Allocation / constraint
Project owner	Scope decisions, label/content review, provider/cloud approvals, official repository access and submission. Capacity is not estimated from five nominal names.
Coding assistant	Drafting, implementation support, automation and technical checks; not independent human evaluation.
Local workstation and repository	Development, offline checks and versioned evidence. Services are started only when needed.
Model allowance	Existing explicitly authorized caps only; each new paid evaluation needs a bounded allocation.
AWS resources	Owner's 7 October screenshot shows a Free-plan account with $100 credits and a zero out-of-pocket constraint. Service access, complete costs and provisioning remain CAP-04 decisions.

Resource	Allocation / constraint
External academic inputs	Lecturer approval/invitation owner-confirmed on 16 October; exact approval reference and numerical rubric remain unprovided. Official upload still needs recording.

KPIs and evidence rules

KPI: Browser task success

Definition and acceptance basis: Successful defined task outcomes / attempted cases in the recorded suite; failures and retries retained.

Current evidence / next measurement: Historical Ticket 15 runs: 43/44, 44/44, 44/44. No new live run claimed here.

KPI: Safety-case compliance

Definition and acceptance basis: Passed designated safety cases / attempted designated cases; all required cases must pass.

Current evidence / next measurement: Historical 19/19 per full run plus release regression evidence; repeat required cases for cloud acceptance.

KPI: Intent and Constraint quality

Definition and acceptance basis: Exact agreement on reviewed targets; class/language support and ambiguity exclusions reported. Existing qualification target is at least 95% with 100% schema/currency/safety compliance on the defined gate corpus.

Current evidence / next measurement: Exploratory intent-only result: 23/53 selected validation correct; synthetic/exposed data, zero unseen. This does not meet or replace full qualification.

KPI: Response/task latency

Definition and acceptance basis: Record p50/p95 with sample count; separate model-inclusive task time, browser execution and provider outage time.

Current evidence / next measurement: Preserve existing RESULTS definitions; cloud threshold and workload require CAP-04/07 agreement. No invented uptime or latency SLA.

KPI: Evaluation cost

Definition and acceptance basis: Attributable provider usage/cost for all requests, including failures and repair; unknowns remain unknown.

Current evidence / next measurement: Provider caps and per-experiment authorization; no budget increase implied by this document.

KPI: User usefulness

Definition and acceptance basis: Report actual observations and method; do not turn qualitative reactions into a satisfaction percentage.

Current evidence / next measurement: Five positive reactions and two account navigations in the informal study; no numeric ratings collected.

KPI: Submission readiness

Definition and acceptance basis: Required artifacts present with evidence links; unresolved external inputs listed; official upload commit recorded.

Current evidence / next measurement: Reports prepared; approval and invitation owner-confirmed; official upload commit pending.

Risk and change management

Risk	Mitigation and evidence
Advice invents product qualities or misreads context	Use bounded fresh catalogue facts, explicit uncertainty, semantic evaluation and retained failures; free prose is not guaranteed by JSON validation.
Unexpected or repeated shopping mutation	Enforce observed targets, bounds, stock, Confirmation, stale/duplicate rejection and uncertain-outcome recovery.
Model quotas or costs interrupt evaluation	Set a bounded experiment allowance before execution, preserve attributable usage, and never increase caps or top up implicitly.
Dataset leakage or weak label coverage	Separate exposed regression material from new data; review labels and keep conversation/paraphrase groups together.
Cloud exposure changes local security assumptions	Review secrets, origins, access controls, session isolation and development endpoints before deployment.
Missing repository access, grading details or approval delays submission	Track the known organization submission requirement and unresolved inputs; prepare editable drafts and evidence now.
Scope expands beyond the capstone	Record a concrete need and approved scope change; preserve the Controlled Storefront boundary.

Every scope change must identify the behaviour or evidence it changes and its acceptance implications. Ticket 16's human-study and live-rerun adjustments are examples of explicit decisions, not permission to relax future cloud gates. Preserve original failed measurements and distinguish proposed work, implemented behaviour and verified outcomes.

Progress and contribution records

Use one row per member and reporting period: assigned work, actual artifact/commit, verification, blockers and instructor feedback. Members confirm their own contributions; role titles alone do not establish authorship. The available baseline is the release and its linked evidence, not a completed individual performance report.

Before submission, check contribution records, export conventions, evidence links and the distinction between completed local work, exploratory ML and planned cloud delivery. Record further approval or rubric details only if supplied. Use the submission checklist. Official upload remains to be recorded.

Export references and provenance

Source content dates are preserved. This PDF was exported on 16 October 2026; it is not an approval or official-upload record. Repository links below use the fixed development snapshot 503b7f6d0897f5a413d546952dd91c9c94d1dec4. They identify supporting evidence, not the official submission destination. External bibliography URLs are preserved as supplied and were not re-verified during export.

Editable source: docs/cap01-project-planning.md
SHA-256: 904e3bf9ff22288e20bfbd9b2b218663079b720d531e56abdcf164f9f84f1e13

Linked sources

1. supplied guidelines

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/depi-guideline-alignment.md

2. CAP-01 register

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/cap01-execution.md

3. RESULTS.md

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/RESULTS.md

4. seven-family comparison

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/cap03-model-comparison-results.md

5. executed notebook

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/notebooks/cap03-model-comparison.ipynb

6. the submission repository

https://github.com/nhahub/NHA-5-145

7. submission checklist

https://github.com/Homd11/shopping-copilot/blob/503b7f6d0897f5a413d546952dd91c9c94d1dec4/docs/depi-guideline-alignment.md#readiness-checklist